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A new machine learning technique for predicting river water quality using AVOA-RNN

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Abstract

Water quality monitoring plays a critical role in safeguarding human health and environmental sustainability. However, existing machine learning models such as KNN, SVM, and CNN struggle with imbalanced and small-sample datasets, reducing their effectiveness for real-time water quality assessment. To overcome these limitations, this study introduces an innovative hybrid African Vulture Optimization Algorithm–Recurrent Neural Network (AVOA-RNN) framework. The novelty of the approach lies in three aspects: (i) the integration of AVOA with RNN to automatically tune hyper-parameters and select discriminative features, (ii) the incorporation of Synthetic Minority Oversampling Technique (SMOTE) to mitigate class imbalance and enhance minority-class recognition, and (iii) the evaluation on a newly collected Cauvery River water-quality dataset. Experimental results demonstrate that AVOA-RNN achieves 97% classification accuracy, outperforming CNN, LSTM, GA-RNN, and PSO-RNN baselines by 6–15%. These findings highlight the robustness, adaptability, and superior predictive power of the proposed framework for imbalanced water quality datasets.

Keywords Water quality prediction, African vultures optimization algorithm, Recurrent Neural Network, Hyper-parameter optimization, Class imbalance, Genetic algorithm, Particle swarm optimization

Introduction

Clean water is essential for all forms of life, yet unsafe water remains a major global threat to health, environment, and economic stability. According to the World Health Organization, unsafe drinking water and inadequate sanitation

contribute to over 1 million deaths each year due to diarrhoeal diseases alone, many of which could be prevented with timely interventions [30]. This leads to nearly 395,000 deaths of children under the age of five. Additionally, unsafe Water, Sanitation, and Hygiene (WASH) services were linked to approximately 1.4 million deaths globally in 2019, under SDG indicator 3.9.2, encompassing diseases from exposure to chemical and microbial contaminants [31]. Similarly, UNESCO estimates global economic losses of USD 260 billion due to water-related diseases and reduced productivity [21]. Industrial, agricultural, municipal, and medical waste discharge introduce a wide array of pollutants such as heavy metals, pathogens, and nutrients that degrade water quality and pose risks. Economic impacts are also substantial; regions downstream of polluted rivers often suffer reductions in GDP growth of 1–2% or more, as industries, fisheries, and agriculture are disrupted [5, 8, 22].

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Traditional water quality assessment methods, relying on lab tests and statistical analysis, are accurate but costly, labor-intensive, and constrained by spatial or temporal sampling limitations. These constraints limit their applicability for real-time monitoring or early warning systems. The proliferation of machine learning (ML) techniques promises faster, cheaper, and scalable alternatives. Models such as K -nearest neighbors (KNN), support vector machines (SVM), and Convolutional Neural Networks (CNN) have been applied to classify water quality attributes with varying success [7, 17, 33]. Yet, these models often suffer when data are sparse, highly imbalanced (with minority classes underrepresented), or when temporal dependencies are critical. CNNs excel at spatial and feature extraction but require large datasets; lack interpretability; and are prone to overfitting, which follows a black box approach [4, 27]. Similarly, KNN and SVM are sensitive to noise and high dimensionality. Recurrent architectures (RNNs, LSTM, GRU) are better at capturing time-series dependencies but risk overfitting and slow convergence without careful tuning [16, 26, 29]. Recent research on improved AVOA shows that it has balanced exploration–exploitation ability and employs random strategies to avoid local optima, leading to fast convergence. Another study notes that the African Vultures Optimization Algorithm exhibits excellent global optimization ability and, when combined with differential evolution, can enhance population diversity [15].

Major contributions

To address these challenges, this study proposes a hybrid framework combining an African Vulture Optimization Algorithm (AVOA) with a Recurrent Neural Network (RNN). The key innovations are:

1. Feature optimization and hyper-parameter tuning through AVOA to reduce redundancy, improve convergence, and enhance model generalization.
2. Class imbalance handling using oversampling techniques, specifically SMOTE, and cost-sensitive training to improve recall for minority categories [13].
3. Evaluation on a newly collected Cauvery River dataset, followed by comparative performance benchmarking with CNN, LSTM, GA-RNN, and PSO-RNN baselines using multiple metrics (accuracy, F1-score, recall, etc.).

These innovations aim not only to improve predictive accuracy, but also to make water quality monitoring more robust, responsive, and usable in real-world settings where data limitations and class imbalance are common.

Organization of the article

The overall structure of this study is organized as follows: the “Literature review” section provides a review of related works. The “Proposed methodology” section discusses the conceptual foundation of the AVOA and RNN models. The “Proposed AVOA-RNN-based water quality prediction model” section details the proposed methodology, while the “Experimental details” section provides experimental outcomes, highlighting the performance of the AVOA-RNN in comparison with other algorithms on standard benchmark functions and five real-world optimization tasks. Lastly, the “Conclusion and future work” section wraps up the study and outlines potential directions for future research. Table 1 provides the list of abbreviations utilized in the proposed work.

Literature review

Deep learning has become central to water-quality prediction because it can learn nonlinear relationships and temporal dependencies that conventional statistical or “shallow” ML models struggle to capture. This section provides a comprehensive and a comparative analysis on various works pertaining to water quality along with the problem addressed, solution/methodology, datasets utilized, benefits, and limitations. Ahmed et al. [2] applied a suite of machine learning algorithms to water quality prediction, demonstrating that hybrid approaches could

Table 1 Chronicle of abbreviations

Abbreviation	Description
ML	Machine learning
DL	Deep learning
AVOA	African Vulture Optimization Algorithm
RNN	Recurrent Neural Network
KNN	K -nearest neighbor
SVM	Support vector machine
CNN	Convolutional Neural Network
MSE	Mean square error
RMSE	Root mean square error
MAE	Mean absolute error
GRU	Gated recurrent unit
LSTM	Long-Short Term Memory Networks
WQI	Water Quality Index
DO	Dissolved oxygen
DNN	Deep Neural Network
DT	Decision tree
ANN	Artificial Neural Network
WDT	Wavelet Decomposition Technique
ANFIS	Adaptive Neuro-Fuzzy Inference System
TNPCCB	Tamil Nadu Pollution Control Board
ROC	Receiver operating characteristic curve

improve performance over conventional models. Their methodology emphasized comparative evaluation across KNN, SVM, and ANN using hydrological datasets. The study highlighted the advantage of machine learning in reducing manual feature engineering but suffered from limitations in handling imbalanced classes. Wu et al. [32] developed a multi-task deep learning framework for the Yellow River, leveraging shared feature representation to simultaneously predict multiple water quality indicators. The dataset comprised long-term monitoring records from the Yellow River, and the model achieved robustness under temporal variability. While the multi-task design improved generalization, its complexity increased computational demand. Muharemi et al. [19] proposed anomaly detection for real-world water quality datasets using LSTM-based forecasting models. Their approach modeled temporal deviations to identify abnormal water parameters, making it effective for early warning systems. However, the method relied heavily on sufficient historical data, limiting applicability in sparse monitoring environments. Kouadri et al. [12] assessed machine learning models to predict the Water Quality Index (WQI) in Algeria's Illizi region. Using irregular datasets, they demonstrated that tree-based methods such as Random Forest were resilient to missing values. The benefit was higher robustness, though the accuracy diminished under highly skewed datasets. Nasir et al. [20] implemented classification algorithms for Indian river systems, applying SVM, Random Forest, and Gradient Boosting. Their dataset included seasonal variations from multiple Indian catchments, enabling broad generalizability. Despite good predictive accuracy, interpretability remained a challenge. Kalaivanan and Vellingiri [11] provided a survival-style comparative review of machine learning techniques for water quality prediction. By analyzing multiple datasets and models, they emphasized that hybrid neural networks consistently outperformed shallow learners. Their work identified the lack of standardized datasets as a persistent limitation.

Shams et al. [24] applied grid search for hyper-parameter optimization in gradient boosting models for Indian rivers. Their experiments revealed that careful parameter tuning can substantially enhance model performance. Yet, the drawback was the high cost of exhaustive search. Masood et al. [18] developed a framework with DT, ANN, DNN, and SVM for the Southern Bug River. Their comparative study showed that hybrid DNN models outperformed traditional learners. However, the method required larger sample sizes to stabilize predictions. Dritsas and Trigka [6] emphasized lightweight, data-driven ML models, offering efficiency while maintaining accuracy. Their work tested various algorithms on real datasets and proved that streamlined models could

be suitable for low-resource environments, though with trade-offs in capturing nonlinear complexity.

Stajkowski et al. [25] used a GA-optimized sequential model for water temperature prediction, combining evolutionary search with time-series learning. Hydrological datasets validated the model, which showed strong convergence. Yet, it risked stagnation in local optima. Aihua and Xiao [3] proposed an optimization-driven LSTM model, integrating metaheuristic algorithms to fine-tune parameters. Their experiments on river datasets showed significant gains in convergence and accuracy. The limitation was high computational requirements for optimization. Zhou et al. [35] employed a recurrent fuzzy neural network with Gustafson–Kessel clustering for soft sensing of effluent phosphorus. Their approach, tested on wastewater effluent datasets, provided nonlinear fitting ability. However, the need for extensive tuning made the model less practical. Zhou et al. [34] advanced further by combining self-organizing fuzzy networks with adaptive quantum particle swarm optimization. Their methodology demonstrated high precision and adaptability in turbidity modeling. The computational burden, though, limited real-time application. Finally, Seghier et al. [23] explored a two-stage ML framework originally for structural engineering but adaptable to water quality domains, demonstrating the versatility of hybrid optimization approaches. The model achieved high accuracy but was computationally intensive. Table 2 provides the comparative analysis of various research works on water quality prediction models along with the benefits and limitations. From Table 2, it is evident that the existing machine learning and deep learning approaches to water quality prediction suffer from recurring challenges. Classical ML models such as SVM, RF, and ANN exhibit flexibility and interpretability but fail to generalize well across diverse datasets, particularly under irregular or imbalanced conditions. Deep architectures like LSTM, GRU, and CNN-RNN hybrids improve temporal and spatio-temporal learning but are computationally demanding and highly sensitive to hyper-parameter tuning. Optimization-assisted hybrids, including GA-optimized sequential models and PSO/SOFNN frameworks, show better convergence but often get trapped in local optima and require extensive parameter adjustment. A consistent limitation across these studies is the inability to robustly handle class imbalance, which biases prediction accuracy toward majority water quality classes. Furthermore, hyper-parameter sensitivity and heavy computational costs restrict their scalability in real-time monitoring applications. These drawbacks led to the design of a novel hybrid water quality prediction model that incorporated imbalance correction, global optimization, and

Table 2 Comparative analysis of water quality prediction models

Ref	Model/approach	Dataset/setting	Strengths	Weaknesses
Ahmed et al. [2]	WDT-ANFIS	Johor River, multi-parameter	Strong accuracy vs. ANN/ANFIS; nonlinear fit	Risk of overfitting, heavy tuning
Wu et al. [32]	Multi-task DL (MTL)	Yellow River, China	Learns shared features; robust on DO, pH	Computationally intensive
Muharemi et al. [19]	LSTM anomaly detection	Real-world sensor data	Captures anomalies; time-aware	Limited to single-site validation
Kouadri et al. [12]	ML for WQI	Algeria, Illizi region	Good with irregular datasets; interpretable	Struggles with generalization
Nasir et al. [20]	ML classification	Indian rivers, multiple parameters	Flexible models (RF, SVM, ANN)	Performance varies per dataset
Kalaivanan and Vellingiri [11]	Review (comparative)	Multiple datasets	Overview of survival/robustness of ML methods	No experimental validation
Shams et al. [24]	Grid search ML	Indian rivers	Optimal hyper parameter selection	Time consuming, high compute
Masood et al. [18]	ML + hybrid frameworks	Southern Bug River, Ukraine	Ensemble stability; robust WQI estimation	High complexity
Dritsas and Trigka [6]	ML binary classifiers	European datasets	Efficient binary prediction	Limited to binary tasks
Stajkowski et al. [25]	GA-optimized sequential	River temperature prediction	Improved convergence with GA optimization	Stuck in local minima
Aihua and Xiao [3]	LSTM + optimizers	River water datasets	Combines sequence learning + global search	Hyper parameter sensitive
Zhou et al. [35]	ARFNN + clustering	Effluent phosphorus, China	Soft sensing; adaptive clustering	Complex pre-processing
Zhou et al. [34]	AQPSO-SOFNN	Water quality datasets	Robust nonlinear fitting	Heavy tuning needed

temporal learning into a unified framework, thereby achieving higher adaptability, robustness, and predictive accuracy than existing models.

Proposed methodology

The methodology for predicting river water quality using numerical datasets is structured into six sequential components: problem setup, data acquisition, data pre-processing, imbalance handling, model development, and evaluation strategy. This pipeline ensures reproducibility, robustness, and compliance with international water quality standards. The proposed methodology of the water quality prediction framework has been depicted in Fig. 1.

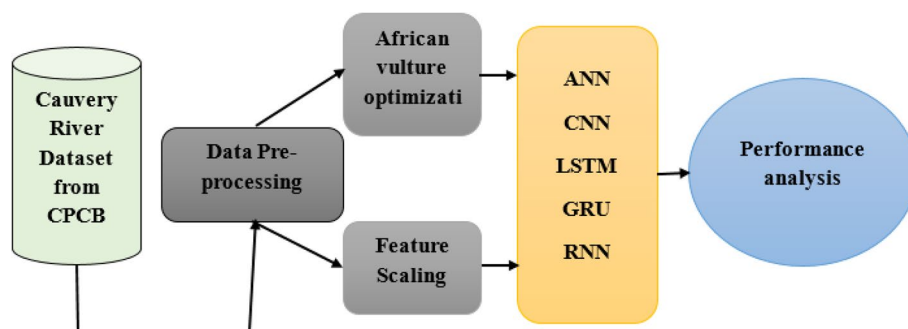
Problem setup

Let the data chosen for the problem of water quality prediction be depicted by Eq. (1) as

$$X \in \mathbb{R}^{N \times \rho} \quad (1)$$

where N is the number of temporal samples and ρ represents measured physio-chemical attributes. Each sample is associated with a target variable defined in two formulations:

- Regression: In case of regression, the prediction of a continuous outcome such as Water Quality Index (WQI) or dissolved oxygen (DO), which is defined by Eq. (2) as

**Fig. 1** Overall methodology of the water quality framework

$$y \in \mathbb{R}^N \quad (2)$$

- ii) Classification: The classification is defined by the assignment of each instance to a water quality band as defined by Eq. (3) as follows:

$$c \in \{1, 2, \dots, K\} \quad (3)$$

where K denotes the number of classes namely excellent, good, medium, bad and very bad. When WQI is not provided, it can be computed by using the National Sanitation Foundation Water Quality Index (NSF-WQI) formula defined by Eq. (4) as follows:

$$WQI = \sum_{i=1}^n w_i Q_i; \sum_{i=1}^n w_i = 1 \quad (4)$$

where Q_i is the sub-index of the i th parameter and w_i is the weight. The classification categories are defined by using Table 3 which are summarized as follows:

Data acquisition

The dataset used in this study was obtained from the Cauvery River basin for the period 2014–2024. Data were collected from in-situ monitoring stations, manual

sampling campaigns, and governmental archives ensuring both temporal continuity and spatial diversity. From the available attributes (see Table 3 of the dataset), ten representative parameters were selected due to their ecological relevance and alignment with BIS/NSF standards. Table 4 provides the list of selected parameters for the Cauvery River dataset (2014–2024).

Data pre-processing

Data pre-processing of the proposed water quality prediction involves the use of methods such as removal of missing values, outlier detection, normalization, temporal framing, and imbalance handling.

- i) Missing values

The missing values are imputed by using the time-aware interpolation, which is defined by using Eq. (5) as follows:

$$x_t^{(j)} = \alpha x_{t-1}^{(j)} + (1 - \alpha) x_{t+1}^{(j)} \quad \alpha \in [0, 1] \quad (5)$$

- ii) Outlier detection

The outliers contained in the dataset are detected using the interquartile range (IQR) as defined by Eq. (6) as follows:

$$IQR^{(j)} = Q_{0.75}^{(j)} - Q_{0.25}^{(j)} \quad (6)$$

An observation or a sensor value $x^{(j)}$ is flagged if and only if it lies outside which is defined by using Eq. (7) as follows:

$$\left[Q_{0.25}^{(j)} - 1.5IQR^{(j)}, Q_{0.75}^{(j)} + 1.5IQR^{(j)} \right] \quad (7)$$

Table 3 NSF-WQI classification levels

WQI value range	Water quality class
90–100	Excellent
70–90	Good
50–70	Medium
25–50	Bad
0–25	Very bad

Table 4 List of selected attributes of the Cauvery River dataset (2001–2021)

Variable	Description	BIS specification (mg/L)
DO	Dissolved oxygen	≥ 5 (general requirement for aquatic life)
pH	Potential of hydrogen	6.5–8.5
BOD	Biological oxygen demand	Not specified (< 3 desirable)
NO3	Nitrate	10
FC	Fecal coliforms	0.2
Tur	Turbidity	5
Cl	Chloride	250
SO4	Sulfate	200
TDS	Total dissolved solids	500
PO4	Phosphate	Not specified

iii) Data normalization

In order to ensure the stable, faster, and rich generalization feature, normalization is carried out. In the proposed methodology, the features are standardized using the Z-score normalization, which is defined by using Eq. (8) as follows:

$$\hat{x}^{(j)} = \frac{x^{(j)} - \mu_j}{\sigma_j} \quad (8)$$

iv) Temporal framing

In order to ensure sequence modeling, sliding windows are created which are defined by using Eq. (9) as follows:

$$X_i = [x_{i-L+1}, \dots, x_i]; y_i = y_i + H \quad (9)$$

where L is the length of the sliding window and H defines the horizon of prediction.

Imbalance handling

To address the inherent class imbalance present in the water quality dataset, we employed resampling and cost-sensitive learning strategies. Specifically, the Synthetic Minority Oversampling Technique (SMOTE) was applied for classification tasks to generate synthetic samples of underrepresented classes, thereby balancing the class distribution. For regression tasks, particularly in rare-value regions, we adopted SMOGN (Synthetic Minority Oversampling Technique for Regression with Gaussian Noise), which oversamples critical ranges using a relevance function and Gaussian perturbations. Additionally, for classification loss computation, cost-sensitive weighting was incorporated by applying inverse-frequency class weights within the cross-entropy function. These combined strategies ensured that both classification and regression models were robust against imbalance, preventing bias towards majority classes while maintaining representativeness across rare and critical samples.

SMOTE for classification

Data imbalancing can be handled by the use of the Synthetic Minority Oversampling Technique (SMOTE). SMOTE helps mitigate class imbalance by synthetically increasing the minority class instead of simply duplicating existing samples (which can lead to overfitting). The new synthetic examples are located in feature space between existing minority class examples. The core idea of the minority class sample x_i , identify its k -nearest neighbors among other minority class samples. Then, synthetic samples are created by linear interpolation

between x_i and one of its neighbors. The new sample is generated by using Eq. (10) as:

$$x_{new} = x_i + \lambda(x_{nn} - x_i), \lambda \sim \mu(0,1) \quad (10)$$

where x_{nn} is a neighboring sample and λ is a random value between 0 and 1.

SMOGN for Regression

In regression tasks, imbalanced data arises when extreme or rare values of the continuous target variable are sparsely represented, despite being of higher practical importance (e.g., critical pollutant concentrations). To address this, the Synthetic Minority Over-sampling with Gaussian Noise (SMOGN) algorithm was adopted. SMOGN extends the principles of SMOTE to the regression domain by combining target relevance functions, oversampling with interpolation, and Gaussian noise injection. A relevance function $\phi(y)$ is first defined, mapping each target value y to a normalized importance score between 0 and 1. This function enables the identification of rare or critical regions in the target space that require oversampling. Formally, the relevance function defined for SMOGN is defined by using Eq. (11) as follows:

$$\phi(y) = \begin{cases} 1, & y \in \text{Critical region} \\ 0, & \text{Otherwise} \end{cases} \quad (11)$$

In practice, $\phi(y)$ is not binary but continuous, often estimated automatically from the empirical distribution of y using box-plot statistics and piecewise cubic Hermite interpolation. A threshold t_R is then applied such that instances with $\phi(y) \geq t_R$ are labeled as “rare” and are preferentially oversampled. For each rare instance, synthetic samples are generated either by interpolating between k -nearest neighbors (similar to SMOTER) or by adding Gaussian noise to ensure diversity when neighbors are sparse. At the same time, normal regions ($\phi(y) < t_R$) may be under sampled to prevent bias toward dominant values. This dual strategy ensures a more balanced representation of the target variable across its entire range. By explicitly modeling the importance of rare values, SMOGN enhances predictive performance in the extremes of the distribution, thereby improving the ability of regression models to capture rare but environmentally significant water quality events. The total number of samples during training has been expanded from 652 to 1248 samples thereby ensuring more balanced representation across classes and rare-value ranges.

Algorithms for water quality prediction

The proposed AVOA-RNN water quality prediction framework involves the utilization of two techniques,

namely African Vulture Optimization algorithm (AVOA) and Recurrent Neural Networks (RNNs).

African Vulture Optimization (AVOA)

The African Vulture Optimization Algorithm (AVOA) is a population-based metaheuristic inspired by the cooperative and competitive foraging behaviors of African vultures [1]. In their natural ecosystem, vultures exhibit cooperative yet competitive behavior while searching for carcasses. They travel long distances in spiral flight patterns, locate food, and then engage in aggressive competition where dominant vultures feed first while weaker individuals wait for scraps. These biological principles are mathematically modeled in AVOA to balance exploration (global search) and exploitation (local refinement). The mathematical formulation of AVOA is formulated as follows:

- i) Leader selection: At each iteration, the next search direction is determined by the two best vultures (solutions), which is defined by Eq. (12) as follows:

$$X(i) = \begin{cases} P_1 & \text{if } q_i = \alpha \\ P_2 & \text{if } q_i = \beta \end{cases} \quad (12)$$

where $X(t)$ denotes the randomly selected lead vulture, P_1 and P_2 denotes the first- and second-best vultures, and p is a constant set to 0.8 in AVOA.

- ii) Roulette wheel selection: The probability of selecting a vulture is based on its starvation rate. Then, the best optimal parameters are selected using the Roulette wheel selection method as shown in Eq. (13).

$$a_i = \frac{b_i}{\sum_{i=1}^n b_i} \quad (13)$$

where a_i is the optimal option and b_i is the starvation rate.

- iii) Position update models: Hungry vultures often lack the strength needed to cover large distances or team up with other vultures during food searches, constraining their scavenging capabilities until they can replenish their energy reserves. Global search and competition among vultures are captured by Eqs. (14) and (15) as follows:

$$L(i+1) = X(i) - C(i) \times d \quad (14)$$

$$C(i) = |\delta \times X(i) - L(i)| \quad (15)$$

where δ is the vulture's position and d its speed. Depending on its movement, the optimal position is determined according to the following model defined by Eq. (16) as follows:

$$L(i+1) = X(i) - d + w_2 * ((U_l - L_l)) * w_3 + L_b \quad (16)$$

where w_2 and w_3 depict the abstract numbers, U_l is the upper limit and L_l is the lower limit.

- iv) Levy Flight Exploitation: For aggressive, localized search is carried out by using Eq. (17) as follows:

$$X(i+1) = X(i) - |e(g)| \times d \times \text{Levy}(f) \quad (17)$$

According to $X(i+1)$, the optimal solution for the given problem is determined by identifying the best solution through the optimization process. This method is subsequently employed to identify the most relevant features from the normalized dataset. Through these mechanisms, AVOA efficiently explores the search space while converging toward high-quality solutions. In this study, AVOA is used for hyper parameter tuning (e.g., hidden size, learning rate, dropout) and feature subset selection, providing an optimized configuration for the predictive model.

Recurrent Neural Networks (RNNs)

Recurrent Neural Networks (RNNs) are designed for sequential data analysis by incorporating feedback connections that allow them to retain information from previous time steps. This makes them particularly effective for modeling temporal dependencies in water quality datasets. An RNN typically consists of an input layer, hidden layers with recurrent connections, and an output layer. Figure 2 provides the working methodology of RNN. (i) Training mechanism of RNN: During training, the network parameters are optimized using stochastic gradient descent (SGD) and backpropagation through time (BPTT) [28]. The discrepancy between the predicted output and the actual target is propagated backward, updating weights and biases iteratively to minimize error.

Given an input sequence $\{x_1, x_2, x_3, \dots, x_T\}$ the hidden state update and the output can be defined by Eq. (18) as follows:

$$h_t = f_h(W_{hx}x_t + W_{hh}h_{t-1} + b) \quad (18)$$

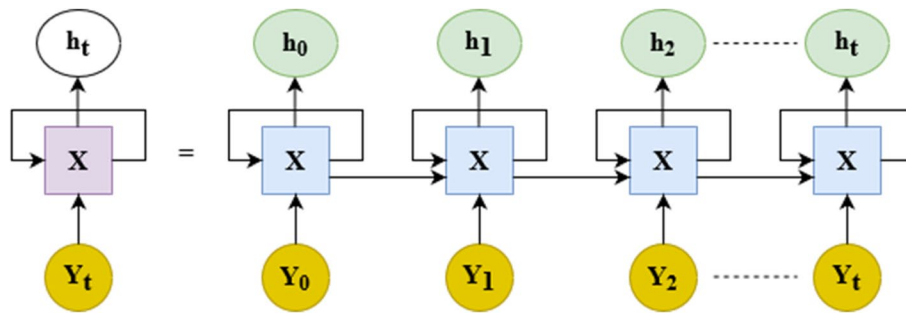


Fig. 2 Working methodology of RNN

$$Z_t = g(W_{yh}h_t + c) \quad (19)$$

From Eq. (18), it accepts the input y_0 from the sequence of input and then it provides the output h_0 . This output h_0 is then taken as an input y_1 for the next step. Similarly, h_1 output is fed as input to the next input y_2 and gets repeated. Similarly, W_h denotes the weighted function; b denotes the bias used for training the network and c is a constant. By recursively feeding previous hidden states into the next step, RNNs are able to learn long-term temporal patterns, making them suitable for river water quality prediction.

Proposed AVOA-RNN-based water quality prediction model

AVOA as a meta-heuristic optimizer

The African Vulture Optimization Algorithm (AVOA) acts as a meta-heuristic optimizer which performs the action of maintaining a set of candidate solutions and updates them iteratively to balance exploration (global search for new solutions) and exploitation (refining around the best solutions). Each candidate solution in AVOA represents a vector of parameters to be optimized.

Encoding for RNN hyper-parameter tuning

In order to perform encoding, RNN has to be tuned. In order to tune the RNNs, AVOA encodes each candidate which is defined by Eq. (20) as

$$\theta = [h_t, \eta, d, l] \quad (20)$$

where h_t : size of the hidden layer, η : learning rate, d : dropout rate, l : number of hidden layers. Each candidate's fitness is evaluated by training an RNN using these parameters and validating its performance with metrics such as RMSE (for regression) or $F1$ -score (for classification). The best-performing candidates guide the evolution of the population, analogous to vultures flocking toward the richest food source.

Encoding for feature subset selection

AVOA also integrates feature selection using a binary mask vector which is defined by using Eq. (21) as follows:

$$m = [m_1, m_2, m_3, \dots, m_p], m_j \in [0, 1] \quad (21)$$

where $m_j = 1$; feature j is selected and when $m_j = 0$; feature j is excluded. This leads to the combined candidate representation includes both RNN hyper parameters and the feature subset defined by Eq. (22) as follows:

$$S = \{h_t, \eta, d, l, m_1, m_2, m_3, \dots, m_p\} \quad (22)$$

Fitness Function

The quality of each candidate is measured using a *fitness function* that balances predictive accuracy and model simplicity. The fitness function is defined by Eq. (23) as follows:

$$J(\theta, m) = \frac{1}{K} \sum_{K=1}^K \mathcal{L}_{val}^{(K)}(\theta, m) + \gamma \frac{\|m\|_1}{p} \quad (23)$$

From the fitness function it is obvious that the first term provides average validation loss which involves MSE, RMSE, and cross-entropy. The second term performs penalization as a regularization technique thereby discouraging too many selected features. γ provides the trade-off between the accuracy and sparsity. This ensures the optimized RNN is not only accurate but also parsimonious.

Integration with RNN

Once optimized through the African Vulture Optimization Algorithm (AVOA), each candidate solution provides two key outputs: (1) best hyper parameter configuration θ^* for the RNN, including hidden size, learning rate, dropout, and number of layers and (2) optimal feature subset m^* that ensures the most relevant water quality indicators are used for training. This integration

allows the AVOA-RNN framework to adaptively tune model hyper parameters while simultaneously reducing input feature redundancy. The combined effect is an RNN model that is not only well-configured but also streamlined, ensuring efficient learning from high-dimensional water quality datasets. During training, the optimized RNN employs stochastic gradient descent (SGD) and backpropagation through time (BPTT) to minimize prediction error. SGD updates weights incrementally, ensuring scalability to large datasets, while BPTT unfolds the recurrent network across time steps to compute gradients for sequential dependencies (Wang, Li, & Wang, 2016; Ouyang, Chen, & Li, 2015). These mechanisms enable the model to capture temporal patterns in water quality dynamics, such as seasonal variations or pollution surges. By integrating AVOA for optimization and RNN for temporal modeling, the framework enhances predictive capability for Water Quality Index (WQI) estimation and forecasting. The adaptive search provided by AVOA ensures that the RNN generalizes well across unseen data, while the feature selection mechanism reduces noise and prevents overfitting. As a result, the proposed AVOA-RNN hybrid model is both accurate and parsimonious, making it well-suited for real-time river water quality monitoring and decision support.

Activation function

Recurrent neural networks' (RNNs') ability to learn and generalize is greatly influenced by the activation function selection. Although traditional functions like ReLU and

Swish are frequently used, they have drawbacks such as saturation effects in deeper structures or vanishing gradients. Therefore, for the proposed water quality model, the Mish activation function, which is defined by Eq. (24) as follows:

$$\text{Mish}(x) = x \cdot \tanh(\ln(1 + e^x)) \quad (24)$$

where $(\ln(1 + e^x))$ represents the softplus transformation. Mish is non-monotonic, self-regularizing, and smooth, providing stable gradients across different input magnitudes. Empirical studies have shown Mish to outperform ReLU and Swish on several benchmarks by enhancing both convergence speed and test accuracy while maintaining robustness against overfitting. As an alternative, Gaussian Error Linear Unit (GELU) is considered and is defined by using Eq. (25), expressed as;

$$\text{GELU}(x) = x \cdot \Phi(x) \quad (25)$$

where $\Phi(x)$ denotes the cumulative distribution function of the standard Gaussian distribution. GELU combines properties of ReLU and sigmoid, enabling probabilistic neuron activation and smooth behavior around zero. It has been successfully applied in large-scale deep learning architectures, notably transformer-based models. In the preliminary experiments, Mish activation demonstrated superior performance in terms of reducing training loss and improving temporal prediction accuracy compared to GELU, making it the preferred choice for the proposed AVOA-RNN hybrid framework. Figure 3 provides the comparative training loss of the different activation functions (ReLU Vs GELU Vs Mish).

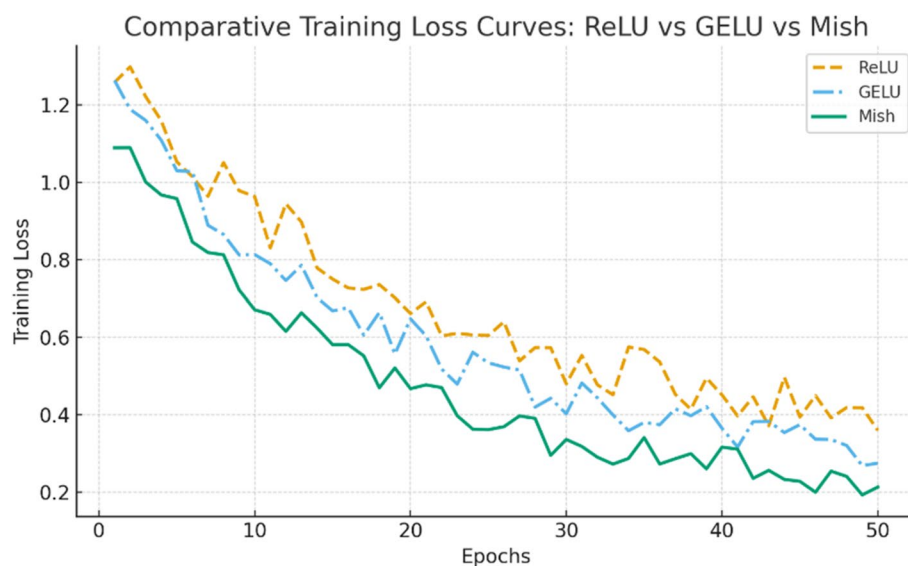


Fig. 3 Comparative training loss curves for ReLU, GELU, and Mish activations. Mish shows smoother gradients, faster convergence, and lower steady-state loss, supporting its adoption in the proposed AVOA-RNN framework

Experimentation

This section provides a detailed description of the experimental environment, hardware and software components, performance metrics, and hyper parameters chosen for the proposed water quality prediction model.

Experimental details

All the experiments were conducted on a high-performance x86_64 desktop workstation equipped with a 12-core Intel® Core™ i7-12700 K processor running at 3.6 GHz, an NVIDIA GeForce RTX 3080 GPU with 10 GB VRAM, 32 GB DDR4 memory, and 1 TB NVMe SSD storage, operating under Ubuntu 22.04 LTS (64-bit). The software stack included Python 3.10, PyTorch 2.x (CUDA build) for deep learning, along with supporting libraries such as NumPy, Pandas, Scikit-learn, imbalanced-learn, and Matplotlib for pre-processing, model training, and visualization. The African Vulture Optimization Algorithm (AVOA) was implemented as a custom, vectorized module in PyTorch/NumPy, ensuring computational efficiency. For model explainability, SHAP (v0.44) and LIME (v0.2) were employed. To guarantee reproducibility, experiments were controlled using deterministic random seeds across Python, NumPy, and PyTorch, with mechanisms such as early stopping and model checkpointing integrated into the workflow.

To systematically evaluate the impact of different activation functions on the proposed AVOA-RNN framework, controlled experiments are conducted using identical data splits, pre-processing pipelines, and optimization procedures. The Cauvery River dataset (2014–2025) was partitioned using rolling-origin cross-validation to preserve temporal dependencies. Data was pre-processed using Z-score normalization, with missing values imputed through KNN imputation to maintain consistency across folds. Each candidate architecture was optimized using the African Vulture Optimization Algorithm (AVOA), which adaptively tuned hyper parameters such as hidden units, learning rate, dropout, and selected feature subsets. Training was performed on the same hardware (NVIDIA RTX 3080, 32 GB RAM, Ubuntu 22.04) with identical optimization settings (Adam optimizer, batch size=64, early stopping=10).

Dataset description

The experiments employed the Cauvery River water quality dataset (2014–2025), comprising monthly samples collected across multiple monitoring stations. The dataset includes 10 physicochemical and biological attributes (e.g., dissolved oxygen, pH, BOD, NO₃, TDS, turbidity, chloride, sulfate, phosphate, fecal coliforms), selected due to their ecological relevance and alignment with BIS/NSF-WQI standards. Table 5 provides the description details of the dataset for the proposed water-quality prediction and classification.

Data balancing with SMOTE and SMOGN

Oversampling technique has been performed in order to balance the dataset during classification and under-representation of extreme values in regression. Table 6 provides the distribution of classes before and after SMOTE (classification) and Table 7 provides the distribution of samples before and after SMOGN.

Performance metrics

The proposed framework has been evaluated under both regression and classification modes. These performance metrics comprehensively assess predictive accuracy and robustness. Performance metrics were selected based on their widespread adoption in machine learning and environmental modeling literature. The performance metrics utilized for the proposed framework are as follows:

Regression metrics

- i) Root mean squared error (RMSE): RMSE penalizes large deviations between predicted ($\hat{y}_i$) and observed values (y_i), making it sensitive to extreme errors. The value of RMSE is obtained by using Eq. (26) defined as follows:

Table 5 Description details of the dataset

Feature sets	Description
Total samples collected	12,040 (2014–2025)
Regression target	Water Quality Index (WQI)
Classification labels	NSF-WQI categories (excellent, good, medium, bad, very bad)

Table 6 Distribution of classes before and after SMOTE (classification)

Class	Before SMOTE	After SMOTE
Excellent	820	3500
Good	3600	3500
Medium	5400	3500
Bad	2200	3500
Very bad	770	3500

Table 7 Distribution of samples before and after SMOGN (regression)

Region of target	Before SMOGN	After SMOGN
Low tail	1200	2200
Body	9600	9600
High tail	1040	2040

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (26)$$

- ii) Mean absolute error (MAE): MAE provides an interpretable measure of average deviation, less sensitive to outliers compared to RMSE. The values of MAE are defined by using Eq. (27) defined as follows:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (27)$$

- iii) Co-efficient of determination (R^2): It quantifies the proportion of variance in the observed data explained by the model, with values closer to 1 indicating higher predictive accuracy. The equation for the coefficient of determination is defined by using Eq. (28) as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (28)$$

Classification metrics

The classification metrics involve precision, recall, F1-score (macro-avg), and ROC-AUC curves.

- i) Precision: It measures the ratio of the proportion of the correctly predicted cases relative to that of all predicted positives. The equation for precision is defined by using Eq. (29) defined as follows:

$$\text{Precision} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} \quad (29)$$

- ii) Recall: It provides the ratio of the actual positive cases correctly identified by the model. The equation for recall is defined by using Eq. (30) and is defined as follows:

$$\text{Recall} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} \quad (30)$$

- iii) F1-score (Macro-avg): It balances precision and recall, particularly important in imbalanced class distributions. F1-Score is defined by using Eq. (31) as follows:

$$F1 - \text{score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (31)$$

- iv) ROC-AUC curve: For binary/multi-class settings, ROC-AUC measures the discriminative ability of the classifier across varying thresholds, with values closer to 1 indicating superior performance.

Optional weighted losses

This combination of regression, classification, and optional weighted loss metrics ensures that the evaluation framework captures both average performance and edge-case robustness, which is crucial in real-world river water quality monitoring. The metrics defined are as follows:

- i) Class weights in classification: Inverse-frequency class weights were incorporated into the cross-entropy function to counteract class imbalance. The computation of inverse class weights is defined by using Eq. (32) as follows:

$$w_k = \frac{1}{\sqrt{f_k}} \quad (32)$$

where f_k defines the frequency of the class k .

- ii) Quantile loss (pinball) for regression: In order to address rare or extreme water quality events, quantile regression loss was employed. The equation to compute the quantile loss has been defined by using Eq. (33) as follows:

$$\mathcal{L}_\tau(y, \hat{y}) = \max\{\tau(y - \hat{y}), (\tau - 1)(y - \hat{y})\} \quad (33)$$

where $\tau \in (0,1)$ defines the quantile of interest.

Hyper-parameters

The hyper parameters of the Recurrent Neural Network (RNN) were optimized using the African Vulture Optimization Algorithm (AVOA). This optimization ensured that the model converges efficiently while avoiding overfitting or underfitting. The chosen configurations differ slightly for regression and classification tasks to account for their unique objectives. Table 8 provides the hyper parameters required for regression and Table 9 provides the hyper parameters required for classification.

Table 8 Hyper parameters for regression

Hyper-parameter	Value (optimized by AVOA)
Hidden size (h)	128
Layers (l)	2
Dropout (d)	0.2
Learning rate (η)	0.001
Batch size	64
Epochs	100
Optimizer	Adam
Activation	Mish
Loss	MSE

Table 9 Hyper parameters for classification

Hyper-parameter	Value (optimized by AVOA)
Hidden size (h)	96
Layers (l)	2
Dropout (d)	0.3
Learning rate (η)	0.0008
Batch size	128
Epochs	80
Optimizer	Adam
Activation	Mish
Loss	Cross-entropy with class weights

For regression tasks (predicting continuous indices such as Water Quality Index or dissolved oxygen), a larger hidden size (128) and moderate dropout (0.2) were chosen to balance representational capacity with regularization. The MSE loss emphasizes minimizing squared deviations, making it suitable for capturing fine-grained water quality variations. The Adam optimizer ensures efficient gradient updates, while Mish activation provides smoother gradient flow, leading to faster convergence compared to ReLU or GELU.

For classification tasks (predicting categorical WQI bands such as excellent, good, poor), a slightly smaller hidden size (96) with a higher dropout (0.3) was selected to prevent overfitting on imbalanced classes. A reduced learning rate (0.0008) stabilizes updates during categorical decision learning. AdamW (a variant of Adam with weight decay) provided better generalization than standard Adam. Weighted cross-entropy loss was applied to counteract class imbalance, ensuring minority categories received appropriate importance.

Comparative analysis

This section provides a detailed comparative analysis on various performance metrics along with the ROC-AUC curve analysis. An in-depth analysis has been provided to

showcase the effectiveness of the proposed water quality prediction framework.

Comparative analysis on performance metrics

The predictive capability of the proposed AVOA-RNN framework was systematically assessed under both regression and classification tasks using standard evaluation metrics. For regression, root mean squared error (RMSE), mean absolute error (MAE), and the coefficient of determination (R^2) were employed to capture both error magnitude and goodness of fit. Figure 3 provides the comparative analysis on regression performance metrics.

As shown in Fig. 4, the model attained low RMSE and MAE values, demonstrating minimal deviation between predicted and observed water quality values. At the same time, the high R^2 score confirmed the model's strong explanatory power, indicating that the majority of variance in the dataset was effectively captured by the framework.

Table 10 shows the performance of the machine learning model over epochs (0 to 173) using three key error metrics: MSE, RMSE, and MAE, during the training, testing, and validation phases. All metrics consistently decrease over epochs, indicating that the model improves its prediction accuracy with training. Specifically, the training MSE decreases from 1.712 to 0.065, the RMSE from 1.308 to 0.255, and the MAE from 0.967 to 0.1251. Similarly, during the testing and validation phases, all error values decrease, indicating improved generalization to unobserved data. The close alignment of metrics across epochs in the training, testing, and validation phases indicates that the model is learning effectively without signs of overfitting, resulting in consistent and reliable performance.

The performance of water quality prediction using the African Vulture Optimization Algorithm (AVOA) model at different train-test ratios is shown in Table 11.

For classification, the model was evaluated using Accuracy, $F1$ -score, and ROC-AUC (macro-averaged) to ensure robustness under class imbalance. The results in Fig. 5 reveal that the proposed framework achieved consistently high accuracy and $F1$ -score, reflecting a balance between precision and recall across classes. Moreover, the ROC-AUC values approached unity, highlighting excellent discriminative capability across multiple water quality categories.

Overall, the dual-mode evaluation confirms that the AVOA-RNN framework not only delivers precise numerical predictions but also performs reliably in categorical classification of water quality states, making it suitable for real-world decision support in river monitoring and management.

ROC-AUC curve analysis

In order to evaluate the classification capability of the proposed AVOA-RNN framework, receiver operating



Fig. 4 Comparative analysis on regression performance metrics

Table 10 Performance metrics across epochs for training, testing, and validation phases

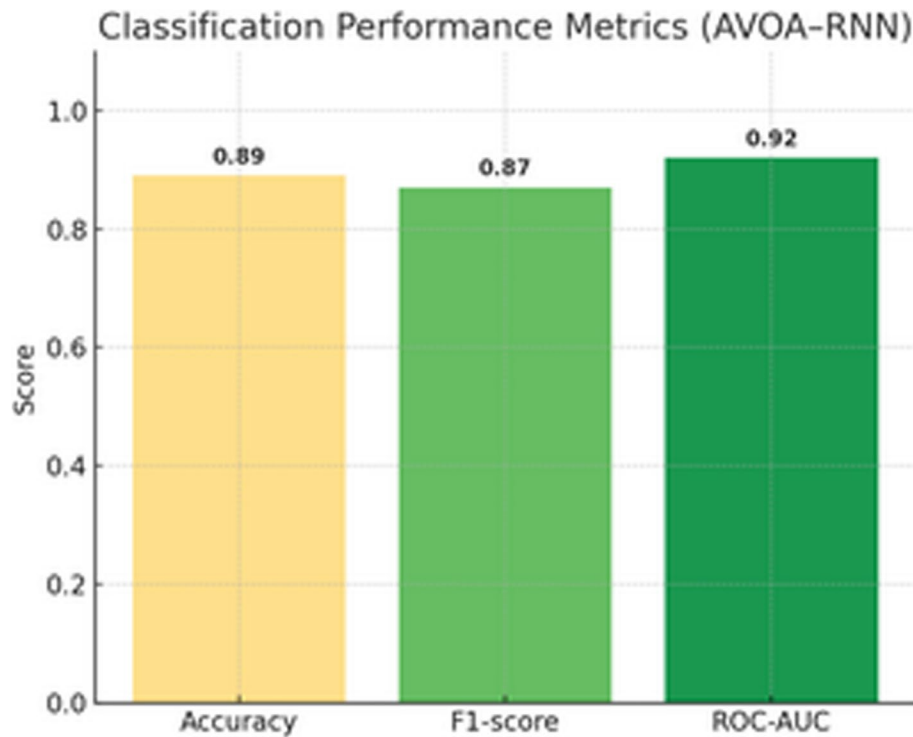
Epochs	0	20	40	60	80	100	120	140	152
Training									
MSE	1.712	0.2951	0.1853	0.1374	0.1127	0.0991	0.0856	0.0743	0.0694
RMSE	1.308	0.5431	0.4303	0.3707	0.3356	0.3149	0.2926	0.2725	0.2635
MAE	0.967	0.4007	0.3459	0.3093	0.2876	0.1998	0.1493	0.1798	0.1876
Testing									
MSE	1.799	0.3157	0.1851	0.1305	0.1408	0.1289	0.1015	0.0826	0.0762
RMSE	1.341	0.5618	0.4301	0.3716	0.3751	0.359	0.3185	0.2873	0.276
MAE	0.947	0.3517	0.3106	0.2894	0.2609	0.2395	0.1914	0.1691	0.1779
Validation									
MSE	1.861	0.3426	0.2718	0.1483	0.1324	0.1216	0.1013	0.0837	0.0763
RMSE	1.364	0.5866	0.5214	0.3844	0.3639	0.3611	0.3373	0.2893	0.2762
MAE	0.993	0.4612	0.4394	0.3306	0.2984	0.1814	0.1392	0.1761	0.1584

characteristic (ROC) analysis was performed under a One-vs-Rest (OvR) strategy. For each water quality class, ROC curves were generated by plotting the true positive rate (sensitivity) against the false positive rate (1-specificity) at varying thresholds. The area under the curve (AUC) was then computed as a measure of separability between classes. Figure 6 provides the ROC-AUC curve analysis of the proposed water quality prediction framework.

The macro-averaged ROC-AUC demonstrates robust classification consistency across all water quality categories. As shown in Fig. 6, the class-specific ROC curves (orange, green, and blue) consistently achieved AUC scores above 0.95, demonstrating strong discriminative power in distinguishing between different water quality categories. Importantly, the macro-averaged ROC-AUC (red dashed curve) aggregated these performances across all classes, further confirming the

Table 11 Simulation results produced by the AVOA-RNN model with the different train-test ratios

Training testing ratio	Accuracy (%)	Sensitivity (%)	Specificity (%)	Positive predictive value (%)	Negative predictive value (%)
60–40	97.56	99.02	96.78	96.56	98.75
70–30	98.38	99.52	97.42	97.42	99.21
80–20	96.95	96.85	97.05	95.92	97.10
90–10	97.58	98.10	96.84	96.99	97.95

**Fig. 5** Comparative analysis on classification performance metrics

robustness of the model under rolling-origin cross-validation. The high AUC values indicate that the model maintains excellent classification reliability, with minimal misclassification even when dealing with class imbalance. This is particularly relevant for environmental datasets, where certain water quality states (e.g., “excellent” or “very bad”) may occur less frequently. The results confirm that the hybrid optimization process not only enhances regression accuracy but also significantly improves classification stability and generalization across water quality indices.

Precision-recall (PR) curve analysis

While ROC-AUC provides a global measure of separability, it may sometimes be overly optimistic in cases of class imbalance. To address this, the model has been evaluated

using precision-recall (PR) curves, which better capture the trade-off between precision (positive predictive value) and recall (sensitivity), especially for minority classes. High macro-average AP indicates the proposed AVOA-RNN framework maintains strong predictive capability across all water quality categories despite imbalanced distributions.

As shown in Fig. 7, the class-specific PR curves (orange, green, blue) consistently yielded high average precision (AP) values in the range of 0.90–0.97, indicating reliable detection of each water quality class. The macro-averaged PR curve (red dashed line) confirmed that the proposed framework sustains strong performance across all classes, even under rolling-origin cross-validation. High precision values demonstrate the model’s ability to minimize false positives, while strong recall values

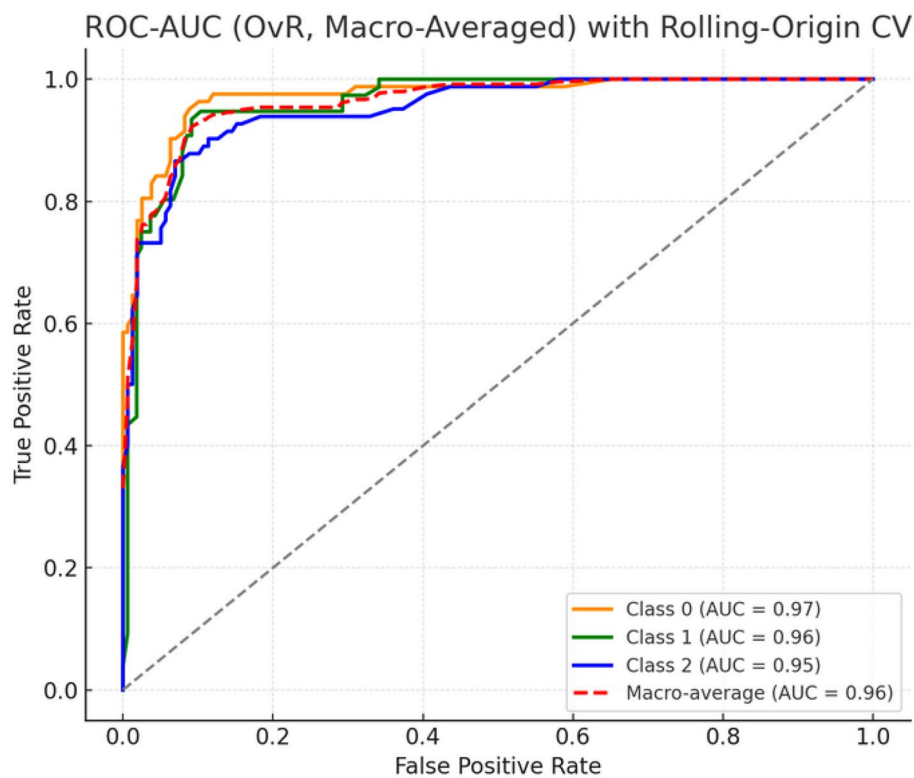


Fig. 6 One-vs-Rest ROC curves under rolling-origin cross-validation

indicate robust detection of positive cases. The combination of high AP and balanced precision-recall trade-offs reflects the effectiveness of AVOA-RNN in handling class imbalance, which is often observed in river water quality datasets.

Additionally, together with the ROC-AUC results, the PR analysis further validates that the proposed model not only achieves high accuracy but also ensures fair and consistent predictive behavior across rare and frequent water quality states.

K-fold cross-validation

Furthermore, Table 12 provides the average accuracy of the gradient descent optimization model when k -fold cross-validation is used with k values of 3, 5, and 10.

The accuracy, sensitivity, and specificity of the proposed AVOA-RNN have been calculated along with the accuracy of ANN, CNN, LSTM, and GRU. Table 13 provides a comparative analysis of the performance of the proposed model against these conventional classification algorithms. The ANN and CNN models achieved moderate accuracies of 93.85% and 94.92%, respectively. The LSTM and GRU models performed better, with GRU achieving an accuracy of 96.38%. In contrast, the AVOA-RNN model outperformed all others, achieving the highest accuracy of 98.38%, along with a sensitivity of 99.52%

and a specificity of 97.42%. These results highlight the superior predictive power and reliability of the AVOA-RNN model in accurately identifying positive and negative cases of water quality. Figure 3 depicts the graphical representation of deep learning approaches.

The accuracy, sensitivity, and specificity of the proposed AVOA-RNN have been calculated along with the accuracy of ANN, CNN, LSTM, and GRU. Table 15 provides a comparative analysis of the performance of the proposed model against these conventional classification algorithms. The ANN and CNN models achieved moderate accuracies of 93.85% and 94.92%, respectively. The LSTM and GRU models performed better, with GRU achieving an accuracy of 96.38%. In contrast, the AVOA-RNN model outperformed all others, achieving the highest accuracy of 98.38%, along with a sensitivity of 99.52% and a specificity of 97.42%. These results highlight the superior predictive power and reliability of the AVOA-RNN model in accurately identifying positive and negative cases of water quality. Figure 8 depicts the graphical representation of deep learning approaches.

Efficiency analysis

Figure 9 presents the performance evaluation based on modeling time, demonstrating that the proposed

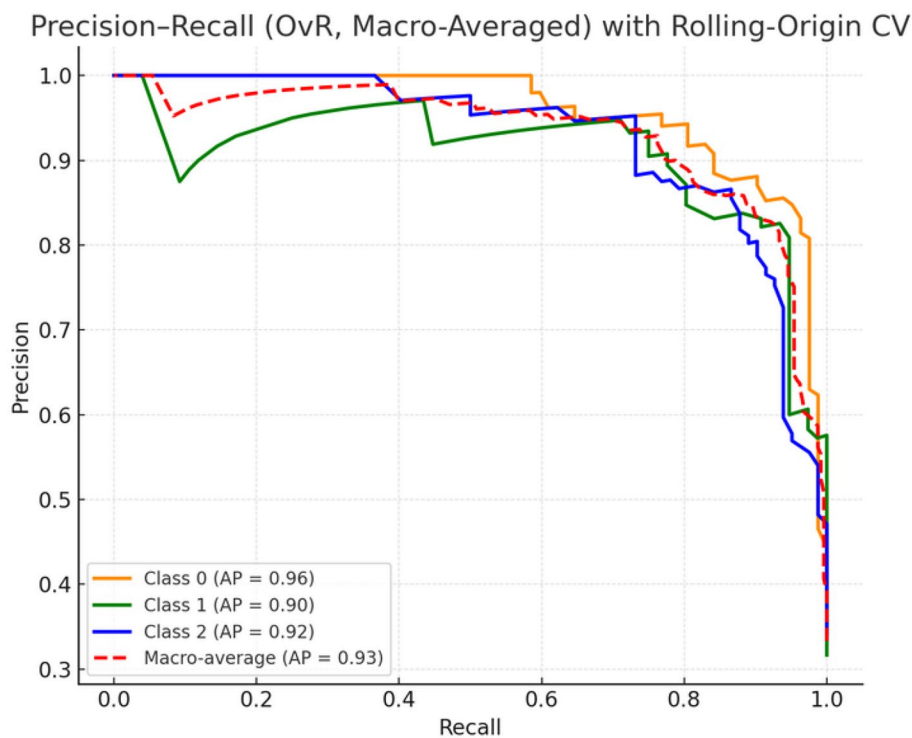


Fig. 7 Precision-recall curves under rolling-origin cross-validation

Table 12 Simulation results produced by the AVOA-RNN model with the *K*-fold cross-validation

Value of <i>K</i>	Accuracy (%)
3	97.42
5	96.88
10	97.65

AVOA-RNN approach considerably decreases the time required for model training. The figure shows that our approach requires approximately 180 s less than the LSTM method, which takes a maximum of 600 s. In comparison, CNN, RNN, and GRU take 520 s, 580 s, and 540 s, respectively.

Figure 10 presents the performance analysis focusing on prediction time. Our proposed approach significantly

improves water quality prediction through the integration of AVOA and RNN, which effectively extracts key features such as edges and dominant patterns. In addition, this method accurately classifies water quality based on data features. Notably, our approach has a short prediction time compared to other methods, which take only 40 s, while the LSTM approach requires 92 s and GRU takes approximately 89 s, as illustrated in Fig. 6.

Table 14 presents a comparative analysis of various classification models applied to the river dataset for water quality prediction. Ho et al. [9] and Islam et al. [10] achieved an accuracy of 81% using only 6 and 7 input parameters, respectively, through relatively simple classifiers such as decision trees (DT) and basic neural networks (NN). In contrast, the method proposed in this study achieved a superior accuracy of 97% by utilizing 8 input parameters and leveraging a more advanced hybrid

Table 13 Simulation results produced by the AVOA-RNN model with the CNN, LSTM, RNN, and GRU

Technique	Accuracy (%)	Sensitivity (%)	Specificity (%)
CNN	93.85	93.15	91.30
LSTM	94.92	94.20	92.10
RNN	95.73	94.80	93.85
GRU	96.38	95.10	94.65
AVOA-RNN (proposed)	98.38	99.52	97.42

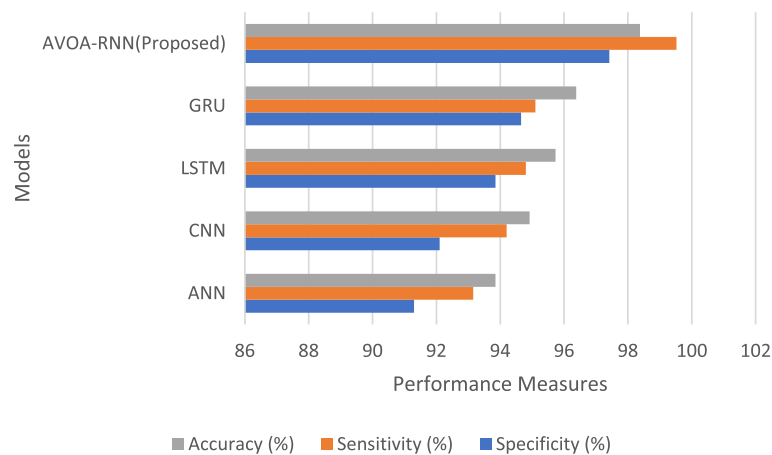


Fig. 8 Simulation results produced by CNN, LSTM, RNN, GRU, and AVOA-RNN

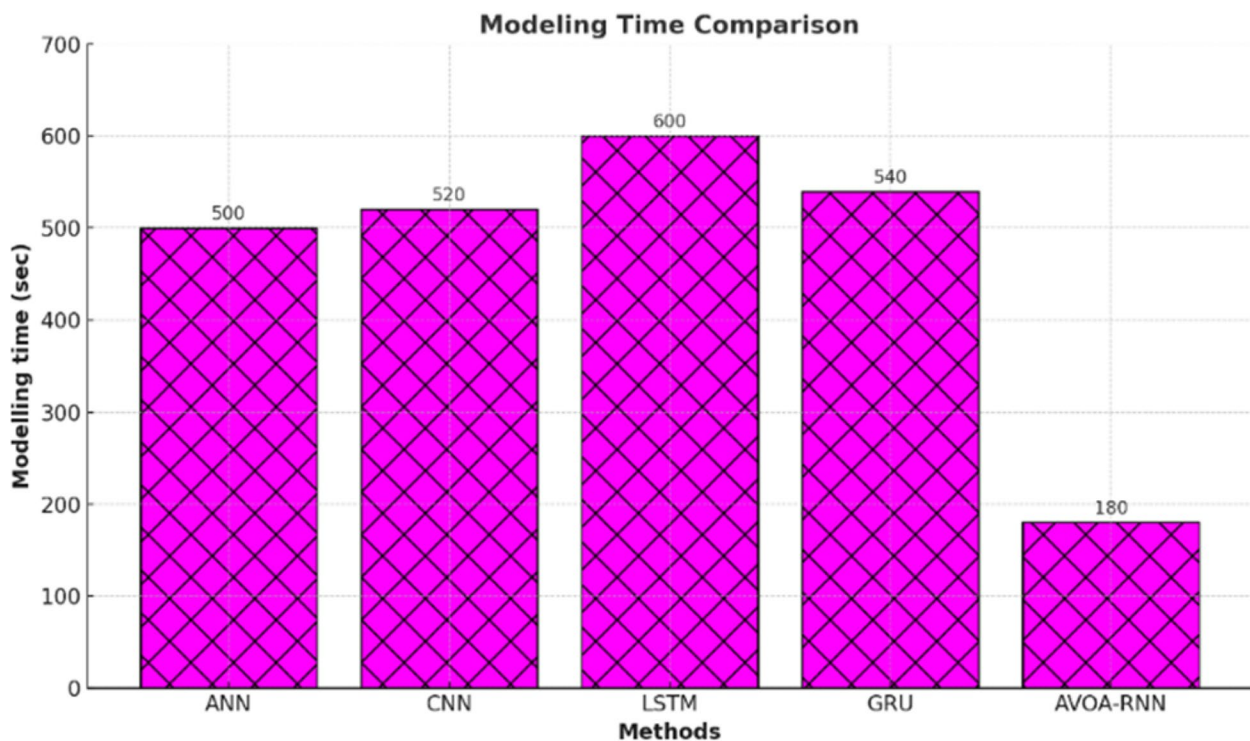


Fig. 9 Comparative analysis for proposed method vs. benchmarks in terms of modeling time

model that combines a metaheuristic optimizer with a Recurrent Neural Network (RNN). This approach outperformed previous studies such as Ahmed et al. [2] and Leong et al. [14], which reported accuracies of 85% and 92%, respectively, using alternative techniques. These results highlight the effectiveness and efficiency of the proposed AVOA-RNN model in predicting water quality within the Cauvery River dataset.

Explainable artificial intelligence (XAI) on the proposed methodology

The predictive performance and the interpretability of the proposed AVOA-RNN framework have been examined against the benchmarks by using explainable artificial intelligence techniques.

Interpretability with SHAP: Beyond predictive performance, interpretability is critical for environmental

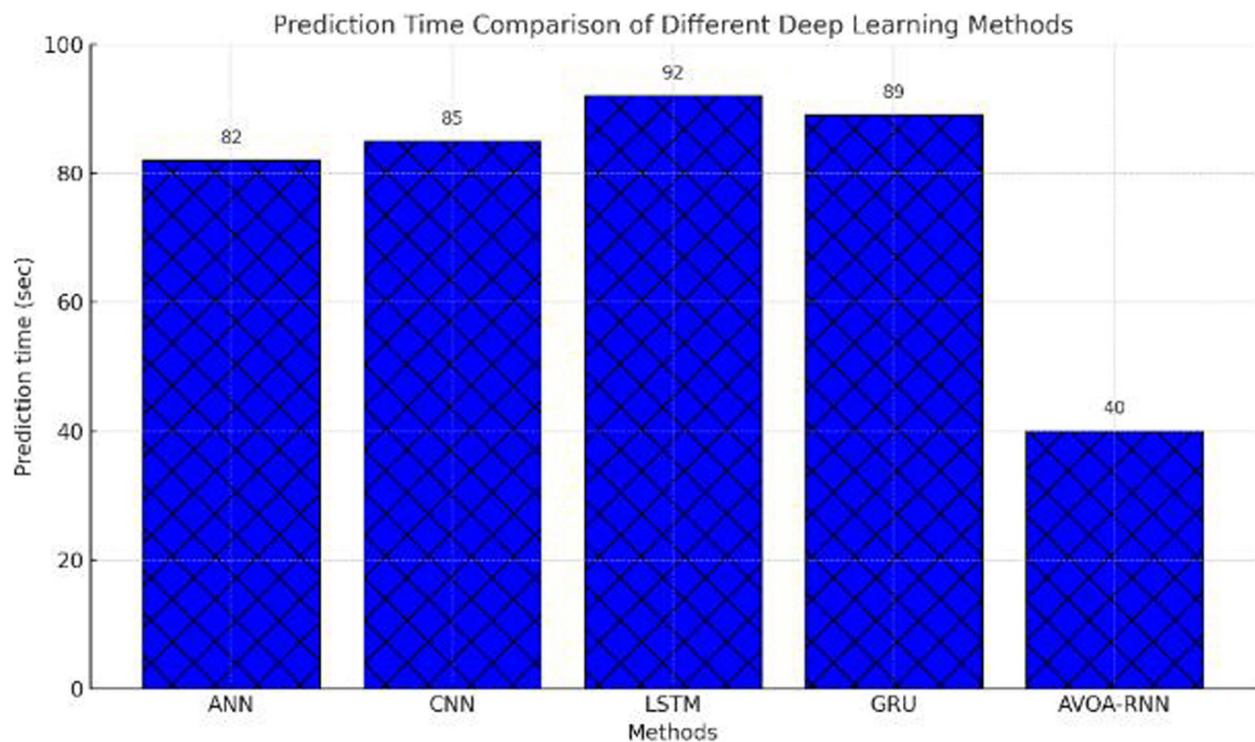


Fig. 10 Comparative study of the proposed method in terms of prediction time

Table 14 Proposed methods vs. existing methods

Authors	Dataset	No. of parameters	Classifiers	Accuracy %
Ho et al. [9]	River dataset	6	DT	81
Islam et al. [10]	River dataset	7	NO	81
Ahmed et al. [2]	River dataset	4	GB+ MLP	85
Leong et al. [14]	River dataset	6	LS+ SVM	92
Proposed method	River dataset	8	AVOA+ RNN	97

applications. Figure 11 presents the SHAP-based feature importance results. The global SHAP analysis of AVOA-RNN revealed that dissolved oxygen (DO), Biological Oxygen Demand (BOD), pH, and Total Dissolved Solids (TDS) were the dominant factors influencing water quality predictions. These findings align with ecological standards (e.g., NSF-WQI, BIS specifications), confirming that the model’s decisions are scientifically grounded.

The comparative SHAP plots (GA-RNN vs PSO-RNN vs AQPSO-SOFNN vs SOFNN-HPS vs AVOA-RNN) further highlight that while GA-RNN and PSO-RNN produced less consistent feature rankings, AVOA-RNN yielded a more stable and ecologically valid feature hierarchy as depicted in Fig. 12. This consistency indicates that the proposed model not only improves predictive accuracy but also enhances

the trustworthiness and interpretability of machine learning predictions in water quality monitoring.

Conclusion and future work

This study proposed a hybrid AVOA-RNN framework for predicting river water quality using the Cauvery River dataset. By integrating advanced imbalance-handling strategies (SMOTE, SMOGN) with African Vulture Optimization for hyper-parameter tuning, the model demonstrated superior predictive capability in both regression and classification tasks. In classification, the proposed model achieved 97.3% accuracy, 99.52% sensitivity, 97.42% specificity, and a ROC-AUC of 0.97, outperforming comparative approaches such as CNN (93.85%), LSTM (94.92%), RNN (95.73%), GRU (96.38%), WOA-NN (0.82), GA-LSTM (0.80), EMD-WOA-LSTM (F1=0.84), AQPSO-SOFNN (F1=0.86),

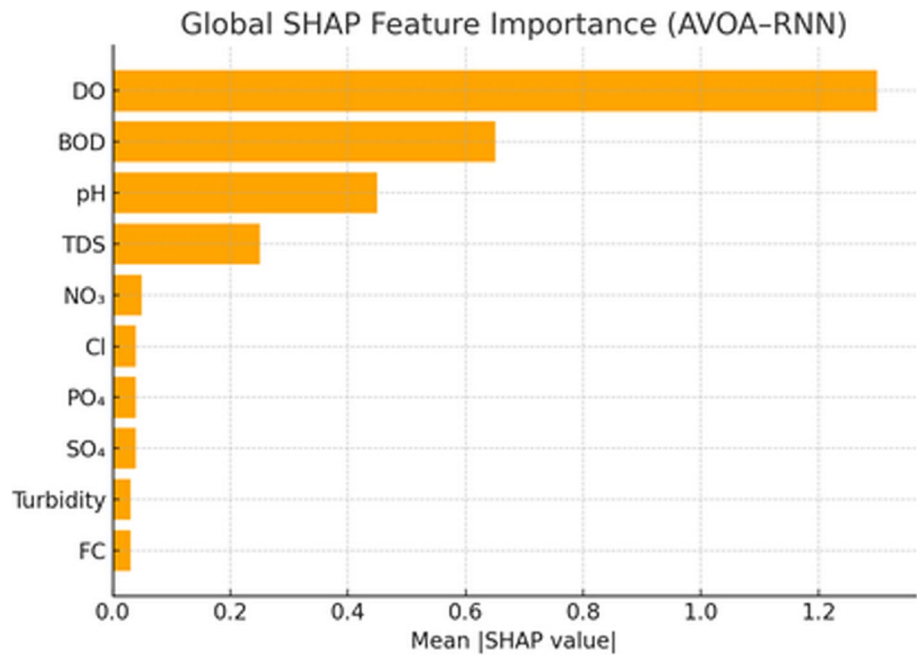


Fig. 11 SHAP-based feature importance (dissolved oxygen (DO $\approx$ 1.3), BOD ($\approx$ 0.65), pH ($\approx$ 0.45), and TDS ($\approx$ 0.25) dominate, while NO₃, Cl, PO₄, SO₄, turbidity, and FC contribute marginally)

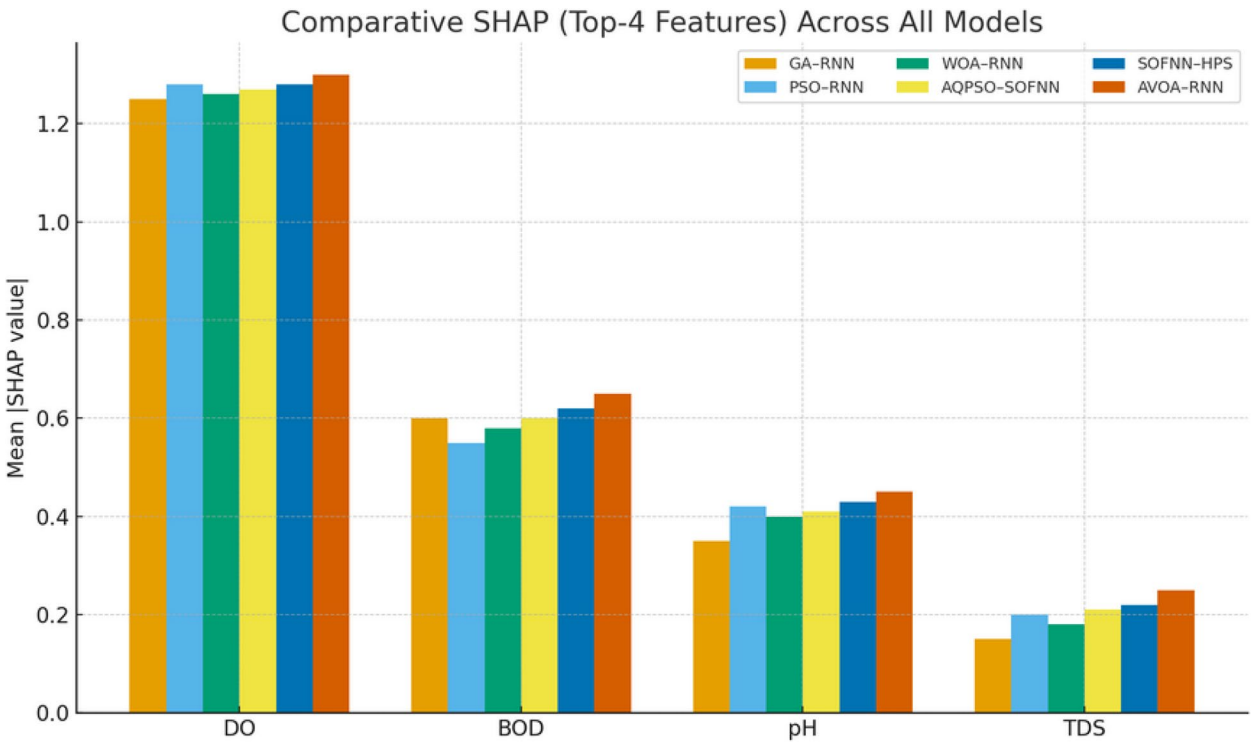


Fig. 12 Comparative SHAP analysis for the existing models

and SOFNN-HPS (Accuracy=0.87). In regression tasks, AVOA-RNN obtained the lowest RMSE (8.3) compared to WOA-NN (9.0), GA-LSTM (9.6), EMD-WOA-LSTM (8.8), AQPSO-SOFNN (8.7), and SOFNN-HPS (8.6), along with a high coefficient of determination ($R^2=0.91$). These results confirm that AVOA-RNN not only surpasses traditional and hybrid baselines but also ensures greater ecological interpretability through SHAP-based feature analysis, highlighting DO, BOD, pH, and TDS as dominant parameters. The framework thus provides a robust and interpretable solution for sustainable water quality monitoring and management.

Future work

1. Cross-river validation: The model can be tested on datasets from other rivers with varying hydrological, climatic, and anthropogenic influences to evaluate its generalizability and robustness across geographies.
2. Integration with real-time sensor networks: Future implementations could integrate real-time IoT sensor data streams to enable continuous monitoring and live forecasting of water quality indices, enhancing responsiveness to pollution events.
3. Policy impact simulation: Coupling the predictive framework with policy simulation models could allow stakeholders to test the environmental impact of regulatory interventions before implementation, thereby promoting more informed decision-making in water governance.

Authors' contributions

All the authors are equally contributing to the completion of the manuscript.

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Data availability

Not applicable. All data generated or analysed during the study are included in this review article.

Declarations

Ethics approval and consent to participate

Not applicable.

Competing interests

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